

Differential equations

$$F(x, y(x), y'(x), y''(x), \dots, y^{(n)}(x)) = 0$$

()th order

implicit form

1st order:

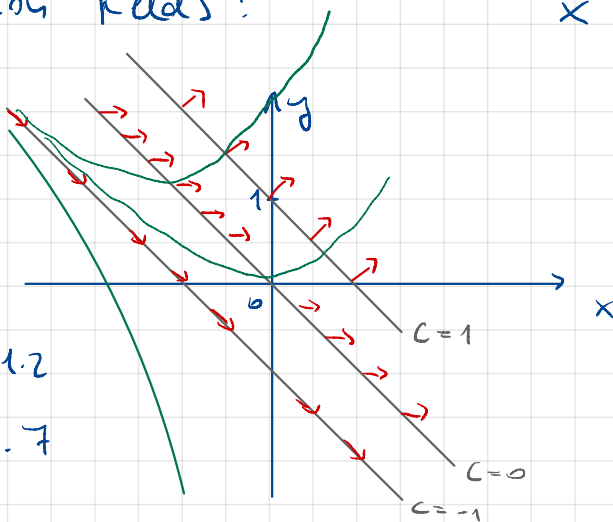
$$F(x, y(x), y'(x)) = 0$$

implicit

$$y' = f(x, y(x))$$

explicit

Direction fields:



Kreyzig 1.2
Fig. 7

initial value

$$x + y(x) - y'(x) = 0$$

$$y'(x) = y(x) + x = f(x, y)$$

↓ slope of solution $y(x)$!

$$f(x, y) = \text{const.} \Rightarrow y = -x + C$$

"C"

$$C=0: \quad y = -x \quad \rightarrow$$

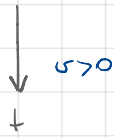
$$C=1: \quad y = -x + 1 \quad \nearrow$$

$$C=-1 \quad y = -x - 1 \quad \searrow$$

$$C=1,5 \quad y = -x + 1,5 \quad \nearrow$$

Dropped ball

$$\dot{v} = g - kv^2$$



$$(k) = 1/\text{length}$$

$$\bar{v} = \sqrt{g/k} \quad \text{terminal velocity}$$

For $v_0 > 0$:

$$v(t) = \bar{v} \cdot \frac{1 - Ce^{-2k\bar{v}t}}{1 + Ce^{-2k\bar{v}t}} \quad \text{where} \quad C = \frac{\bar{v} - v_0}{\bar{v} + v_0}$$

Check:

$$\text{LHS: } \dot{v} = \bar{v} \cdot \left[-Ce^{-2k\bar{v}t} \cdot (-2k\bar{v}) (1 + Ce^{-2k\bar{v}t}) - (1 - Ce^{-2k\bar{v}t}) (Ce^{-2k\bar{v}t} (-2k\bar{v})) \right] \\ \hline (1 + Ce^{-2k\bar{v}t})^2$$

$$= \frac{\bar{v} \cdot 2k\bar{v}C e^{-2k\bar{v}t} \cdot 2}{(1 + Ce^{-2k\bar{v}t})^2}$$

$$\text{RHS: } g - kv^2 = g - k\bar{v}^2 \frac{(1 - Ce^{-2k\bar{v}t})^2}{(1 + Ce^{-2k\bar{v}t})^2} =$$

$$= \frac{g(1 + 2Ce^{-2k\bar{v}t} + C^2 e^{-4k\bar{v}t}) - \bar{v}^2(1 - 2Ce^{-2k\bar{v}t} + C^2 e^{-4k\bar{v}t})}{(1 + \dots)^2}$$

$$= \frac{4gCe^{-2k\bar{v}t}}{(\dots)^2} = \frac{4k\bar{v}^2Ce^{-2k\bar{v}t}}{(1 + Ce^{-2k\bar{v}t})^2} \quad \checkmark \quad \text{OK}$$

$$v(0) = \bar{v} \frac{1-C}{1+C} = \bar{v} \frac{1 - \frac{\bar{v}-v_0}{\bar{v}+v_0}}{1 + \frac{\bar{v}-v_0}{\bar{v}+v_0}} = \bar{v} \frac{2v_0}{2\bar{v}} = v_0 \quad \checkmark \quad \text{OK}$$

• $t \gg \frac{1}{k\bar{v}}$: $e^{-2k\bar{v}t} \ll 1$

use $\frac{1}{1+x} = 1-x + O(x^2)$:

$$\begin{aligned} v(t) &= \bar{v} (1 - Ce^{-2k\bar{v}t}) (1 - Ce^{-2k\bar{v}t} + O(e^{-4k\bar{v}t})) \\ &= \bar{v} [1 - 2Ce^{-2k\bar{v}t} + O(e^{-4k\bar{v}t})] \end{aligned}$$

• $t \ll \frac{1}{k\bar{v}}$: use $e^{-x} = 1 - x + O(x^2)$

$$v(t) \approx \bar{v} \frac{1 - C(1 - 2k\bar{v}t)}{1 + C(1 - 2k\bar{v}t)} = \frac{\bar{v} \frac{1-C}{1+C} + \frac{C}{1+C} \cdot 2k\bar{v}^2 t}{1 - \frac{C \cdot 2k\bar{v}t}{1+C}}$$

$$\approx \left(v_0 + \frac{C}{1+C} 2k\bar{v}^2 t \right) \left(1 + \frac{C}{1+C} 2k\bar{v}t \right)$$

$$= v_0 + (v_0 + \bar{v}) \frac{C}{1+C} 2k\bar{v}t + O(t^2)$$

$$= v_0 + (\bar{v}^2 - v_0^2) kt + O(t^2) = v_0 + gt - kv_0^2 t + O(t^2)$$

• $v_0 \gg \bar{v} = \sqrt{g/k}$

$$\rightarrow \dot{v}(t) \approx -k v(t)^2$$

$$v(t) = \frac{v_0}{1 + kv_0 t}$$

$$v(0) = v_0 \quad \checkmark$$

$$\dot{v}(t) = - \frac{v_0}{(1 + kv_0 t)^2} kv_0 = -k v(t)^2 \quad \checkmark$$

$t \ll \frac{1}{kv_0}$: $v(t) \approx v_0 (1 - kv_0 t)$

Kreyzig 1.1 Problems

1.) $y' + 2 \sin(2\pi x) = 0$

$$y = \frac{1}{\pi} \cos(2\pi x) + C$$

2.) $y' + x e^{-x^2/2} = 0$

$$y = e^{-x^2/2} + C$$

3.) $y' = y$

$$y = A \cdot e^x$$

6.) $y'' = -y$

$$y = A \cdot \sin x + B \cdot \cos x$$

or

$$y = A' e^{ix} + B' e^{-ix}$$

12.)

$$y \cdot y' = 4x$$

$$y^2 - 4x^2 = C \quad (y > 0)$$

$\downarrow \frac{d}{dx}$

$$2yy' - 8x = 0 \quad \checkmark$$

$$y(1) = 4$$

$\downarrow$

$$4^2 - 4 \cdot 1^2 = C$$

$$C = 12$$

or

$$y = \sqrt{C + 4x^2}$$

$$y' = \frac{8x}{2\sqrt{C+4x^2}}$$

$\downarrow$

$$y \cdot y' = 4x \quad \checkmark$$

10.) $y' + 5xy = 0$

$$y = C e^{-2.5x^2}$$

$$y(0) = \pi$$

$\downarrow$

$$y' = C \cdot e^{-2.5x^2} \cdot (-2.5 \cdot 2 \cdot x) = -5x \cdot y \quad \checkmark$$

$$\pi = C \cdot e^0 = C$$

14)

$$y' \cdot \tan x = 2y - 8$$

$$y = C \sin^2 x + 4$$

$$y(\pi/2) = 0$$

$$y' = 2C \sin x \cos x$$

$$y' \cdot \tan x = 2C \sin^2 x = 2y - 8 \quad \checkmark$$

$$0 = C \cdot \sin^2(\pi/2) + 4 =$$

$$= C + 4 \rightarrow C = -4$$

18.) Radium $T = 3.6 \text{ days} = t_{1/2}$

$$\dot{y} = -k \cdot y \rightarrow y = y_0 \cdot e^{-k \cdot t}$$

$$\frac{y_0}{2} = y_0 \cdot e^{-k \cdot T} \rightarrow -k \cdot T = \ln \frac{1}{2}$$

$$T = \frac{\ln 2}{k} \rightarrow k = \frac{\ln 2}{T} = 0.1925 \frac{1}{\text{day}}$$

a) $y_0 = 1g, t = 1 \text{ day}:$

$$y = 1g \cdot e^{-0.1925 \cdot 1} \approx 0.825g$$

$$y = y_0 e^{-k \cdot t} = y_0 (e^{-k \cdot T})^{t/T} = y_0 (1/2)^{t/T}$$

b)

$$t = 365 \text{ day}:$$

$$y = 1g \cdot e^{-365 \cdot 0.1925} = 1g \cdot \left(\frac{1}{2}\right)^{86.3,6} = 1g \cdot 2^{-100}$$

$$2^{10} \approx 10^3 \rightarrow \approx 10^{-30}g \ll m_{\text{proton}}!$$